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HIGH VOLTAGE SOLID INSULATED CAPACITIVE-RESISTIVE DIVIDER

Abstract

A capacitive voltage divider designed to step down high-voltage from an electrical power line to a lower more suitable voltage includes a hollow insulator, a primary terminal located on the hollow insulator and connected to the high-voltage power line, first and second capacitors arranged as a voltage divider circuit, one or more primary insertion resistors, and solid dry foam dielectric formed around the first capacitor and the second capacitor within the hollow insulator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority under 35 U.S.C. 102(e) to U.S. Provisional Application No. 63/560,207 filed Mar. 1, 2024, the contents of which are hereby incorporated by reference in their entirety.

FIELD

[0002] The present disclosure relates to voltage dividers, and in particular, to resistive capacitive voltage dividers used in the electrical power and distribution industry.

BACKGROUND

[0003] The electrical power grid needs to be monitored at all times to facilitate the safe and continuous supply of electricity. To this end, the ability to detect and respond to problems is paramount. Monitoring requires obtaining current and voltage measurements and using these measurements for monitoring, metering, relaying and remediation throughout the system. As will be appreciated, electrical transmission lines operate at high voltages. These high voltages pose a risk to auxiliary equipment and secondary equipment such as Intelligent Electronic Devices (IED). Accordingly, utility companies have a need to convert high transmission class voltages or step down voltages into lower voltages that may be used for management functions such as measurement, monitoring, relaying control and metering. The lower voltages are also used to control protective relays in the power grid. Successful conversion of high voltage (HV) to a representative low voltage (LV) for use in the management of the power grid is essential.

[0004] A high voltage (HV) voltage divider is a device used to step down high transmission line voltages into the lower voltages that are representative and more useful in managing and monitoring the electrical grid. A capacitive voltage transformer (CVT) is a common and cost-effective type of device used by electric utility companies to step down high transmission line voltages to lower voltages. A capacitor coupled voltage transformer (CCVT) is similar to a CVT, but with the addition of a carrier coupling unit that allows the CCVT to couple power line carrier (PLC) signals to a transmission line for communication purposes. The terms CVT and CCVT are often used interchangeably.

[0005] FIG. 1 depicts a simplified circuit diagram of a conventional CVT **100**. CVT **100** includes capacitors **111** and **113** which are surrounded by a liquid or gaseous dielectric material **101**, and the transformer **103**. Due to the high voltages being managed, the capacitors in these conventional devices are placed in either porcelain or composite hollow insulators that are filled with a liquid dielectric (e.g., mineral oil) or a gaseous dielectric (e.g., SF₆ gas). One drawback of such conventional designs is that they contain environmentally hazardous materials that cause problem if the CVT breaks or wears out and needs to be disposed of. Another drawback is that these conventional CVT voltage dividers may be flammable or even explosible. Moreover, voltage dividers that are filled with gaseous dielectrics may be categorized as pressure vessels, thus requiring compliance with American Society of Engineers (ASME) code. Another drawback in using gaseous SF₆ as a dielectric is that it is one of the most potent greenhouse gases on earth, and thus, quite environmentally unfriendly.

[0006] Most conventional CVT designs rely on capacitive condensers which tend to be prone to accuracy drift with increasing age. Another drawback of convention CVT dividers is presented by the ever-growing need to accurately assess power quality in the modern electrical grid. Assessing power quality requires measuring all of the harmonic levels in the system. Conventional CVTs of the type described above are unable to account for harmonics in the system without the use of peripheral equipment. Accordingly, there remains a need in the art for a CVT that measures the harmonics that conventional CVTs cannot.

SUMMARY

[0007] Various embodiments disclosed herein that are advantageous over the conventional voltage dividers currently in use in the power transmission and distribution industry. For example, various embodiment may be implemented as a newly designed geometry graded electrode systems

containing a solid dielectric material. Various embodiments of the solid dielectric are advantageous in that they deter leakages and typically do not require replenishment of the dielectric media. Some embodiments may be implemented with nonflammable (and nonexplosive) types of solid dielectric material. The various embodiments implemented as a straightforward geometry grading system are relatively inexpensive to manufacture. The various embodiments are also versatile inasmuch as they may be expandable as a capacitive-resistive dividers (CR dividers). The embodiments are also advantageous in that the dry solid foam or solid gel dielectric material flows in and around the internal components of the voltage divider and then solidifies, enhancing structural support.

[0008] Various embodiments provide a capacitive voltage divider apparatus configured to step down a high-voltage from an electrical power line to a low-voltage. The apparatus includes solid dielectric material, a hollow insulator with a distal end and a proximal end, the hollow insulator containing at least some of the solid dielectric material. The apparatus also includes a primary terminal positioned at the distal end of the hollow insulator, the primary terminal being configured for an electrical connection to a conductor at the high-voltage of the electrical power line. A high voltage electrode of the apparatus is within and surrounded by the solid dielectric material. The apparatus also includes a high voltage lead that electrically connects the primary terminal to the high voltage electrode. The high voltage lead is at least partially within the solid dielectric material. The apparatus also includes voltage divider circuitry within the solid dielectric material and a shield component included as part of the voltage divider circuitry. The shield component is in proximity to the high voltage electrode, and the shield component is within and surrounded by the solid dielectric material.

[0009] In some embodiments of the capacitive voltage divider apparatus include first and second portions of the solid dielectric material. The capacitive voltage divider may also include a component container affixed to the proximal end of the hollow insulator. The voltage divider circuitry and second portion of the solid dielectric material may be contained within the component container. In some embodiments of the capacitive voltage divider the component container is a CVT tank, and the voltage divider circuitry and dielectric material are contained within the CVT tank. In such embodiments the voltage divider circuitry is within and surrounded by the second portion of the solid dielectric material.

[0010] Some embodiments of the capacitive voltage divider apparatus include first and second capacitors positioned within and surrounded by the second portion of the solid dielectric material. The first capacitor includes a first lead and a second lead, and the second capacitor includes a third lead and a fourth lead. The second lead is electrically connected to the third lead. In some embodiments of the capacitive voltage divider apparatus the voltage divider circuitry is an axial grading capacitive voltage divider.

[0011] In some embodiments of the capacitive voltage divider apparatus the shield component is a calotte shield with the first lead of the first capacitor is electrically connected to the calotte shield and the fourth lead of the second capacitor is connected to a ground potential. Some embodiments include a terminal box which has a first terminal electrically connected to the second lead of the first capacitor and a ground terminal electrically connected to ground potential.

[0012] In some embodiments the high voltage lead includes a first high voltage lead section with a first end and a second end. The first end is electrically connected to the primary terminal. The high voltage lead may have a second high voltage lead section with a third end and a fourth end.

[0013] Some embodiments include a primary insertion resistor that is electrically connected between the second end of the first high voltage lead section and the third end of the second high voltage lead section. The fourth end of the second high voltage lead section may be electrically connected to the high voltage electrode. In various embodiments the high-voltage is at least 7,000 volts and the low-voltage is no greater than 500 volts.

[0014] Various embodiments provide a voltage divider apparatus configured to step down a high-voltage from an electrical power line to a low-voltage. The apparatus may include solid dielectric

material, a hollow insulator with a distal end and a proximal end, and containing at least some of the solid dielectric material. The apparatus further includes a primary terminal positioned at the distal end of the hollow insulator. The primary terminal is configured for an electrical connection to a conductor at the high-voltage of the electrical power line. The apparatus also includes a high voltage electrode within and surrounded by the solid dielectric material, and a high voltage lead that electrically connects the primary terminal to the high voltage electrode. The high voltage lead is at least partially within the solid dielectric material. The apparatus also includes voltage divider circuitry within the solid dielectric material, and a cylindrical shield component included as part of the voltage divider circuitry. The cylindrical shield component is in proximity to the high voltage electrode and is within and surrounded by the solid dielectric material.

[0015] In various embodiments of the voltage divider apparatus the solid dielectric material includes a first portion and a second portion. Some embodiments also include a component container affixed to the proximal end of the hollow insulator. In some embodiments the voltage divider circuitry and a second portion of the solid dielectric material are contained within the component container.

[0016] In some embodiments the component container is a CVT tank. The voltage divider circuitry and dielectric material are contained within the CVT tank, within and surrounded by the second portion of the solid dielectric material. In some embodiments the high voltage electrode has a cylindrical section, and the voltage divider circuitry further includes a coaxial ground cylinder electrode electrically connected to ground potential. In some embodiments the high voltage electrode, the cylindrical shield component and the coaxial ground cylinder are concentric. The high voltage electrode is positioned within the cylindrical shield component, and the cylindrical shield component is positioned within the coaxial ground cylinder. In various embodiments the voltage divider circuitry is a radial geometry grading capacitive voltage divider. Some embodiments include a terminal box mounted on an outside surface of the component container. A first terminal within the terminal box is electrically connected to the cylindrical shield component, and a ground terminal within the terminal box is electrically connected to the ground potential. In some embodiments the high-voltage is at least 7,000 volts and the low-voltage is no greater than 500 volts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] Various aspects and advantageous features of the present disclosure will become more apparent to those of ordinary skill in the art from the explanations in the Detailed Description below, and reference to the accompanying drawings, wherein:

[0018] FIG. 1 depicts a circuit diagram of a conventional capacitive voltage divider (CVT) filled with liquid or gaseous dielectric material.

[0019] FIGS. 2A-B depict cross-sectional views of axial geometry grading capacitive voltage dividers in accordance with various embodiments disclosed herein.

[0020] FIGS. 2C-D depict cross-sectional views of radial geometry grading capacitive voltage dividers in accordance with various embodiments disclosed herein.

DETAILED DESCRIPTION

[0021] Presently preferred embodiments of the present disclosure illustrating a capacitive voltage divider for use in the power industry is depicted in FIGS. 2A-D. While the description of these embodiments is set forth in connection with a CVT for use in the power industry whether in generation, substations, renewables, or other areas where an LV signal representative of an HV signal is needed, one of ordinary skill in the art will readily appreciate that the advantageous features described herein may be used in any field or environment where an independent

measurement of an HV signal is needed.

[0022] FIGS. 2A and 2B are cross-sectional views depicting implementations **200** and **210** of axial geometry grading capacitive voltage dividers with solid dielectric material **201** in accordance with various embodiments disclosed herein. Use of the solid dielectric material **201** in the various embodiments eliminates the need for gaseous or liquid dielectrics thus overcoming the disadvantages of conventional gaseous or liquid dielectrics—e.g., dielectric leakage, environmental hazards, and in some cases, flammability or explosiveness. Further, the use of a straight-forward electrode-based grading system in the various embodiments avoids the need for capacitive condensers which are a source of inaccuracy in conventional voltage dividers.

[0023] In various embodiments the solid dielectric material **201** may be implemented as a solid dry foam dielectric or may be a solidified gel dielectric. In some of the various embodiments the solid dielectric material **201** may be rigid like extruded polystyrene (e.g., Styrofoam™) or more flexible like a silicone rubber compound that cures into a solidified dielectric silicone gel. Two suitable dielectrics are Wacker SilGel™ and Dow SYLGARD™ 527 solid Silicone Dielectric Gel. In other implementations the solid dielectric material **201** may be flexible like extruded polyethylene (e.g., flexible packing foam sheets).

[0024] Turning to FIG. 2A, the axial grading capacitive voltage divider implementation **200** has a high voltage primary terminal **211** which may be electrically connected to a high voltage source such as a high voltage electrical transmission or distribution line. A high voltage lead **213** runs through a hollow insulator **217**, connecting the high voltage primary terminal **211** to high voltage electrode **215**. The hollow insulator **217** may be made from an insulative composite material, in which case it is called a hollow composite insulator (HCI) **217**. In some implementations the hollow insulator **217** may be porcelain or other such insulating material as are known by those of ordinary skill in the art. The hollow insulator **217** has a distal end (top in FIG. 2A) and a proximal end (bottom). The high voltage lead **213** also runs through concentric inner and outer HCI shields **219** contained within the hollow composite insulator **217**. The inner and outer HCI shields **219** shields aid in preventing the effects of high field gradients within the device due to the high voltages, and also provide shielding of stray capacitances.

[0025] The hollow composite insulator **217** sits on top of CVT tank **221**. The CVT tank **221** is a component container that contains voltage divider circuitry **229**, an axial grading capacitive voltage divider. The CVT tank **221** may be made from aluminum, a composite material, iron, stainless steel, or other such material that is of sufficient sturdiness to support the hollow composite insulator **217**. The high voltage lead **213** extends into CVT tank **221**. A high voltage electrode **215** is at the end of high voltage lead **213**. The CVT tank **221** is typically filled with the solid dielectric material **201**. The dielectric material **201** is flowed into CVT tank **221** as a liquid, and then solidifies into solid dielectric material **201**. This results in the components of voltage divider circuitry **229** being within and surrounded by the solid dielectric material **201**. Once it is solidified the solid dielectric material **201** adheres to the surfaces of the components of voltage divider circuitry **229**. The hollow composite insulator **217** taken together with CVT tank **221** may, in some instances, be considered to be a single container that holds the solid dielectric material **201**. CVT tank **221** of FIG. 2A contains a number of grading shields **223** which are part of the voltage divider circuitry **229**. The grading shields **223** are supported by a supporting insulator **225** affixed to the bottom of CVT tank **221**. A terminal box **227** is connected on the outside of CVT tank **221**. The terminal box **227** contains terminals that are electrically connected to the various grading shields **223**, providing access to the stepped down voltages.

[0026] Conventional CVTs contain dielectrics in liquid or gaseous form such as mineral oil or SF6. The conventional liquid/gaseous dielectrics flow freely within the CVT container, surrounding all the internal components of the voltage divider. The various embodiments have this same advantageous feature since the dielectric material **201** is flowed in (or forced in) in liquid form, surrounding all the internal components of capacitive voltage divider **200**. The dielectric material

201 solidifies as it cools, forming a dry foam (or solid gel) dielectric that surrounds the internal components of capacitive voltage divider **200**. The solid dielectric material **201** of the various embodiments is advantageous over conventional liquid/gaseous dielectrics in that it solidifies around and clings to the internal components, providing some structural support. The added structural support aids in preserving the structural integrity of the capacitive voltage divider **200** as it experiences vibrations and movement due to wind, traffic, trains, earthquakes, or other outside influences.

[0027] FIG. 2B depicts a cross-sectional view of an axial geometry grading capacitive voltage divider implementation **210** in accordance with various embodiments. FIG. 2B is similar to the voltage divider shown in FIG. 2A, but with the addition of high voltage (HV) primary insertion resistors **241-243** and substitution of medium voltage secondary voltage divider circuitry **239**. The axial grading capacitive voltage divider **210** has a high voltage primary terminal **211** which may be electrically connected to a high voltage source such as a high voltage electrical transmission or distribution line. Segments of high voltage lead **213** electrically connect the components of axial grading capacitive voltage divider **210** within the hollow insulator **217**. HV primary insertion resistors **241-243** are connected in series between the high voltage primary terminal **211** and the high voltage electrode **215**. Three HV primary insertion resistors are shown in FIG. 2B. However, the device may be implemented with more or fewer high voltage primary insertion resistors, depending upon the requirements of the installation and availability or cost of components. The insertion of HV primary insertion resistors **241-243** makes axial grading capacitive voltage divider **210** suitable for very high harmonics transients. The addition of the HV primary insertion resistors **241-243** makes the arrangement of the components similar to a Zaengl voltage divider, sometimes called a series-damped capacitive divider. However, prior to the present invention no Zaengl voltage divider design has been implemented with a solid dielectric as are the various embodiments disclosed herein.

[0028] The hollow insulator **217** may be made from an insulative composite material, in which case it is called a hollow composite insulator (HCI) **217**. In some implementations the hollow insulator **217** may be porcelain or other such insulating material as are known by those of ordinary skill in the art. The hollow insulator **217** has a distal end, towards the top of the figure, and a proximal end towards the bottom of the figure. The high voltage lead **213** and/or one or more of the HV primary insertion resistors **241-243** run through concentric inner and outer HCI shields **219** contained within the hollow composite insulator **217**. The inner and outer HCI shields **219** shields aid in preventing the effects of high field gradients within the device due to the high voltages, and also provide shielding of stray capacitances.

[0029] The hollow composite insulator **217** sits on top of CVT tank **221** which contains the medium voltage (MV) secondary voltage divider circuitry **239**. The CVT tank **221** may be made from aluminum, a composite material, iron, stainless steel, or other such material that is of sufficient sturdiness to support the hollow composite insulator **217**. The high voltage lead **213** extends into CVT tank **221**. A high voltage electrode **215** is positioned at the end of high voltage lead **213** in proximity to calotte shield **235**. This results in a voltage V_1 at calotte shield **235**. The high voltage electrode **215** may be spherical in shape, or may be implemented with oval or otherwise rounded cross-section, depending upon the shape of calotte shield **235**. The CVT tank **221** is typically filled with the solid dielectric material **201**. The dielectric material **201** is flowed into CVT tank **221** as a liquid, and then solidifies into solid dielectric material **201**. This results in the components of voltage divider circuitry **239** being within and surrounded by the solid dielectric material **201**. Once it is solidified the solid dielectric material **201** adheres to the surfaces of the components of voltage divider circuitry **239**. The hollow composite insulator **217** taken together with CVT tank **221** may be considered to be a single container that holds the solid dielectric material **201**.

[0030] CVT tank **221** of FIG. 2B contains medium voltage (MV) secondary voltage divider

circuitry **239**. The C1 capacitor **231** of the MV secondary voltage divider circuitry **239** is electrically connected between calotte shield **235** and the C2 capacitor **233**. The other side of C2 capacitor **233** is electrically connected to ground. This design provides a stepped down voltage terminal, and a ground terminal to terminal box **227**. FIG. 2 depicts two capacitors **231/233**. However, in practice the MV secondary voltage divider circuitry **239** may be implemented with more capacitors so as to provide more voltage terminals within terminal box **227** at various stepped down voltages.

[0031] The C1 capacitor **231** and C2 capacitor **233** are in series between the voltage V1 at calotte shield **235** and ground. Capacitors C1 and C2 are shown as two capacitors in the simplified circuit diagram of FIG. 2B. However, in practice C2 (and sometimes C1) may itself be embodied as multiple capacitors in parallel or in a circuit configuration of parallel and/or series capacitors. For example, C2 may consist of multiple capacitors C3+C4+C5 in parallel (not shown).

[0032] The MV secondary voltage divider circuitry **239** steps down the voltage V1 at calotte shield **235** to a useable lower voltage V2 which is routed to the terminal box **227**. The terminal box **227** contains a terminal that is electrically connected to between the capacitors C1 and C2 and another terminal at ground potential, providing access to the stepped down voltage from MV secondary voltage divider circuitry **239**. To step down the relatively high voltage V1 to a lower, useable voltage V2 the capacitor C2 must typically be much larger than capacitor C1. The lower voltage V2 is a ratio of the capacitance values of C1 and C2 applied to the higher voltage V1 as follows:

$$[00001] V2 = V1 \times C1 / (C1 + C2)$$

[0033] The solid dielectric material **201** is shown surrounding both C1 and C2. In some implementations, however, the C1 and C2 capacitors may be contained in different containers or different compartments of the CVT tank **221**. This allows the C1 and C2 capacitors to be packed in different types of the solid dielectric material **201**. Depending upon the expected requirements and conditions of the CVT, the solid dry foam dielectric **101** may be implemented in various degrees of hardness/flexibility.

[0034] FIG. 2C depicts a cross-sectional view of radial geometry grading capacitive voltage divider implementation **220** in accordance with various embodiments disclosed herein. The radial geometry grading capacitive voltage divider **220** has a high voltage primary terminal **261** at voltage V1 which may be electrically connected to a high voltage source such as a high voltage electrical transmission or distribution line. A high voltage lead **263** runs through a hollow insulator **267**, connecting the high voltage primary terminal **261** to high voltage electrode **265** which is also at voltage V1. The high voltage electrode **265** typically is cylindrically shaped, or as in FIG. 2C (and FIG. 2D), has a cylindrically shaped section with tapered ends.

[0035] The hollow insulator **267** may be made from an insulative composite material, or porcelain, or another such insulating material as are known by those of ordinary skill in the art. The hollow insulator **267** has a distal end (top in FIG. 2C) and a proximal end (bottom). The high voltage lead **263** also runs through concentric inner and outer HCI shields **269** contained within the hollow composite insulator **267**. The inner and outer HCI shields **269** shields aid in preventing the effects of high field gradients within the device due to the high voltages, and also provide shielding of stray capacitances.

[0036] The hollow composite insulator **267** sits on top of CVT tank **271**. The CVT tank **271** is a component container that contains the voltage divider circuitry **279**, a radial geometry grading capacitive voltage divider. The CVT tank **271** may be made from aluminum, a composite material, iron, stainless steel, or other such material that is of sufficient sturdiness to support the hollow composite insulator **267**. The high voltage lead **263** extends into CVT tank **271**, connecting to high voltage electrode **265**. The CVT tank **271** is typically filled with solid dielectric material **201**. The dielectric material **201** is flowed into CVT tank **271** as a liquid, and then solidifies into solid dielectric material **201**. This results in the components of voltage divider circuitry **279** being within and surrounded by the solid dielectric material **201**. Once it is solidified the solid dielectric material

201 adheres to the surfaces of the components of voltage divider circuitry **279**. The hollow composite insulator **267** taken together with CVT tank **271** may be considered to be a single container that holds the solid dielectric material **201**. CVT tank **271** contains two grading shields **273** and a ground screening shield **274** (sometimes called guard electrode) which are part of the voltage divider circuitry **279**. The grading shields **273** and ground screening shield **274** are supported by a supporting insulator **275** affixed to the bottom of CVT tank **271**. A terminal box **277** is connected on the outside of CVT tank **271**. The terminal box **277** contains terminals V2, V3 and GND that are electrically connected to the grading shields **273** and the ground screening shield **274**, providing access to the stepped down voltages.

[0037] FIG. 2D depicts a cross-sectional view of radial geometry grading capacitive voltage divider implementation **230** in accordance with various embodiments disclosed herein. The radial geometry grading capacitive voltage divider **230** has a high voltage primary terminal **261** at voltage V1 which may be electrically connected to a high voltage source such as a high voltage electrical transmission or distribution line. Segments of high voltage lead **213** electrically connect the components of radial geometry grading capacitive voltage divider **230** within the hollow insulator **267**. High voltage (HV) primary insertion resistors **241-243** are connected in series between the high voltage primary terminal **211** and the high voltage electrode **265**. Three HV primary insertion resistors are shown in FIG. 2D. However, the device may be implemented with more or fewer high voltage primary insertion resistors, depending upon the requirements of the installation and availability or cost of components. The insertion of HV primary insertion resistors **241-243** makes radial geometry grading capacitive voltage divider **230** suitable for very high harmonics transients. [0038] The hollow insulator **267** has a distal end, towards the top of the figure, and a proximal end towards the bottom of the figure. The high voltage lead **213** and/or one or more of the HV primary insertion resistors **241-243** run through concentric inner and outer HCI shields **219** contained within the hollow composite insulator **267**. The inner and outer HCI shields **219** shields aid in preventing the effects of high field gradients within the device due to the high voltages, and also provide shielding of stray capacitances. The inner and outer HCI shields **269** shields aid in preventing the effects of high field gradients within the device due to the high voltages, and also provide shielding of stray capacitances.

[0039] The hollow composite insulator **267** sits on top of CVT tank **271**. The CVT tank **271** is a component container that contains the components of the secondary voltage divider circuitry **289**, a radial geometry grading capacitive voltage divider. The CVT tank **271** may be made from aluminum, a composite material, iron, stainless steel, or other such material that is of sufficient sturdiness to support the hollow composite insulator **267**. The high voltage lead **263** extends into CVT tank **271**, connecting to high voltage electrode **265**. The CVT tank **271** is typically filled with the solid dielectric material **201**. The dielectric material **201** is flowed into CVT tank **271** as a liquid, and then solidifies into solid dielectric material **201**. This results in the components of voltage divider circuitry **289** being within and surrounded by the solid dielectric material **201**. Once it is solidified the solid dielectric material **201** adheres to the surfaces of the components of voltage divider circuitry **289**. The hollow composite insulator **267** taken together with CVT tank **271** may be considered to be a single container that holds the solid dielectric material **201**. CVT tank **271** contains a HV coaxial cylindrical electrode **265**, a coaxial tap electrode **283** and a coaxial ground cylinder electrode **284** (sometimes called guard electrode) which are part of secondary voltage divider circuitry **289**. The coaxial tap electrode **283** and coaxial ground cylinder electrode **284** are at least partly cylindrical. In some implementations the coaxial tap electrode **283** and coaxial ground cylinder electrode **284** may have a cylindrical center section with flared ends. The HV coaxial cylindrical electrode **265**, coaxial tap electrode **283** and coaxial ground cylinder electrode **284** are supported by a supporting insulator **275** affixed to the bottom of CVT tank **271**. A terminal box **277** is connected on the outside of CVT tank **271**. Terminal box **277** contains terminals V2 and GND that are electrically connected to the coaxial tap electrode **283** and coaxial ground cylinder

electrode **284**, providing access to the stepped down voltages.

[0040] The term “electrically connected” as used herein means that two components are connected by an electrically conductive path, i.e., galvanically connected. In some implementations a component may be electrically connected to another component via one or more other components so that current flows between the components and/or the voltage of one component is affected by the other component. For example, the bottom lead of capacitor **C2** of FIG. **1** may be connected to the grounded terminal of transformer **103**, which in turn, is connected to ground. In another example, the top lead of **C2** (which connects to **C1**) may be connected to transformer **103** via compensating reactor circuitry designed to compensate for the phase shift caused by the capacitive voltage divider formed by **C1** and **C2**. Two components may be “directly electrically connected”, in which case they are connected to each other rather than being connected via a third component.

[0041] As described above, the present disclosure provides embodiments of a capacitive voltage divider that does not pose an environmental risk from the use of gaseous or liquid insulants and/or a risk of shattering from use of a porcelain vessel under pressure. The present disclosure also provides embodiments of a capacitive voltage divider having a simple metallic electrode-based grading system and avoids capacitive condensers that are prone to accuracy drift with increasing age. Further, as described above, various of the above and other embodiments of the present disclosure provide a capacitive voltage divider with improve power quality monitoring.

[0042] One of ordinary skill will appreciate that the exact dimensions and materials are not critical to the disclosure and all suitable variations should be deemed to be within the scope of the disclosure if deemed suitable for carrying out the objects of the disclosure. One of ordinary skill in the art will also readily appreciate that it is well within the ability of the ordinarily skilled artisan to modify one or more of the constituent parts for carrying out the various embodiments of the disclosure. Once armed with the present specification, routine experimentation is all that is needed to determine adjustments and modifications that will carry out the present disclosure.

[0043] In the various embodiments components are said to be within and surrounded by the solid dielectric material. A component that is “within and surrounded by” another material is engulfed by it on all sides. In some cases the dielectric material may adhere to the surfaces of the component, and in other instances it may not. For example, a person underwater in a swimming pool is within and surrounded by the water, but the liquid water does not adhere to the person. The components of a voltage divider are within and surrounded by a solid foam dielectric if the foam dielectric is flowed in and around the components as a liquid and then solidifies. Also, once it is solidified the solid foam dielectric adheres to the voltage divider components. A component wrapped in or layered in solid sheets of dielectric material is not within and surrounded by the sheets of dielectric material.

[0044] The embodiments described above are for illustrative purposes and are not intended to limit the scope of the disclosure or the adaptation of the features described herein to particular optical voltage sensing systems or electro-optic crystal assemblies. Those skilled in the art will also appreciate that various adaptations and modifications of the above-described preferred embodiments can be configured without departing from the scope and spirit of the disclosure. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described.

Claims

1. A capacitive voltage divider apparatus configured to step down a high-voltage from an electrical power line to a low-voltage, the apparatus comprising: solid dielectric material; a hollow insulator with a distal end and a proximal end, the hollow insulator containing at least some of the solid dielectric material; a primary terminal positioned at the distal end of the hollow insulator, the primary terminal being configured for an electrical connection to a conductor at the high-voltage of

the electrical power line; a high voltage electrode within and surrounded by the solid dielectric material; a high voltage lead that electrically connects the primary terminal to the high voltage electrode, the high voltage lead at least partially within the solid dielectric material; voltage divider circuitry within the solid dielectric material; and a shield component included as part of the voltage divider circuitry, the shield component being in proximity to the high voltage electrode, wherein the shield component is within and surrounded by the solid dielectric material.

2. The capacitive voltage divider apparatus of claim 1, wherein the at least some of the solid dielectric material is a first portion of the solid dielectric material, the capacitive voltage divider further comprising: a component container affixed to the proximal end of the hollow insulator; wherein the voltage divider circuitry and a second portion of the solid dielectric material are contained within the component container.

3. The capacitive voltage divider apparatus of claim 2, wherein the component container is a CVT tank; wherein the voltage divider circuitry and dielectric material are contained within the CVT tank; and wherein the voltage divider circuitry is within and surrounded by the second portion of the solid dielectric material.

4. The capacitive voltage divider apparatus of claim 2, wherein the voltage divider circuitry comprises: a first capacitor positioned within and surrounded by the second portion of the solid dielectric material, the first capacitor including a first lead and a second lead; and a second capacitor positioned within and surrounded by the second portion of the solid dielectric material, the second capacitor including a third lead and a fourth lead; wherein the second lead is electrically connected to the third lead.

5. The capacitive voltage divider apparatus of claim 1, wherein the voltage divider circuitry is an axial grading capacitive voltage divider.

6. The capacitive voltage divider apparatus of claim 4, wherein the shield component is a calotte shield; and wherein the first lead of the first capacitor is electrically connected to the calotte shield and the fourth lead of the second capacitor is connected to a ground potential.

7. The capacitive voltage divider apparatus of claim 6, further comprising: a terminal box; a first terminal within the terminal box electrically connected to the second lead of the first capacitor; and a ground terminal within the terminal box electrically connected to the ground potential.

8. The capacitive voltage divider apparatus of claim 1, wherein the high voltage lead comprises a first high voltage lead section with a first end and a second end, the first end being electrically connected to the primary terminal; and wherein the high voltage lead comprises a second high voltage lead section with a third end and a fourth end.

9. The capacitive voltage divider apparatus of claim 8, further comprising: a primary insertion resistor electrically connected between the second end of the first high voltage lead section and the third end of the second high voltage lead section; wherein the fourth end of the second high voltage lead section is electrically connected to the high voltage electrode.

10. The capacitive voltage divider apparatus of claim 9, wherein the high-voltage is at least 7,000 volts and the low-voltage is no greater than 500 volts.

11. A voltage divider apparatus configured to step down a high-voltage from an electrical power line to a low-voltage, the apparatus comprising: solid dielectric material; a hollow insulator with a distal end and a proximal end, the hollow insulator containing at least some of the solid dielectric material; a primary terminal positioned at the distal end of the hollow insulator, the primary terminal being configured for an electrical connection to a conductor at the high-voltage of the electrical power line; a high voltage electrode within and surrounded by the solid dielectric material; a high voltage lead that electrically connects the primary terminal to the high voltage electrode, the high voltage lead at least partially within the solid dielectric material; voltage divider circuitry within the solid dielectric material; and a cylindrical shield component included as part of the voltage divider circuitry, the cylindrical shield component being in proximity to the high voltage electrode, wherein the cylindrical shield component is within and surrounded by the solid

dielectric material.

12. The voltage divider apparatus of claim 11, wherein the at least some of the solid dielectric material is a first portion of the solid dielectric material, the capacitive voltage divider further comprising: a component container affixed to the proximal end of the hollow insulator; wherein the voltage divider circuitry and a second portion of the solid dielectric material are contained within the component container.

13. The voltage divider apparatus of claim 12, wherein the component container is a CVT tank; wherein the voltage divider circuitry and dielectric material are contained within the CVT tank; and wherein the voltage divider circuitry is within and surrounded by the second portion of the solid dielectric material.

14. The voltage divider apparatus of claim 12, wherein the high voltage electrode has a cylindrical section, and wherein the voltage divider circuitry further comprises: a coaxial ground cylinder electrode electrically connected to ground potential.

15. The voltage divider apparatus of claim 14, wherein the high voltage electrode, the cylindrical shield component and the coaxial ground cylinder are concentric.

16. The voltage divider apparatus of claim 15, wherein the high voltage electrode is positioned within the cylindrical shield component, and the cylindrical shield component is positioned within the coaxial ground cylinder.

17. The voltage divider apparatus of claim 16, wherein the voltage divider circuitry is a radial geometry grading capacitive voltage divider.

18. The voltage divider apparatus of claim 17, further comprising: a terminal box mounted on an outside surface of the component container.

19. The voltage divider apparatus of claim 18, further comprising: a first terminal within the terminal box electrically connected to the cylindrical shield component; and a ground terminal within the terminal box electrically connected to the ground potential.

20. The voltage divider apparatus of claim 19, wherein the high-voltage is at least 7,000 volts and the low-voltage is no greater than 500 volts.
